

Dossier : High fat diet syndromes and obesity

Effects of polydatin from *Polygonum cuspidatum* on lipid profile in hyperlipidemic rabbits

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Abstract

Hyperlipidemia is one of the vital coronary risk factors and is positively related to morbidity and mortality of coronary heart disease. There are numerous herbal medicines which are reported to exert good hypolipidemic actions with few side effects.

In the present study, the hypolipidemic effects of polydatin, a compound from *Polygonum cuspidatum* Sieb. et Zucc, on hyperlipidemic rabbits were evaluated. Thirty-two male rabbits were fed a high fat/cholesterol diet for 6 weeks and another eight male rabbits fed a basic diet served as normal control. Three weeks after a high fat/cholesterol diet, the animals were orally administrated polydatin (25, 50, and 100 mg kg⁻¹ per day) by intubation for 3 weeks. The results showed that polydatin markedly decreased the serum levels of total cholesterol (TC), triglyceride (TG), and low-density lipoprotein cholesterol (LDL-C) in hyperlipidemic rabbits. The ratio of TC to high-density lipoprotein cholesterol (HDL-C) and the liver coefficient were also reduced. But both polydatin and high fat/cholesterol diet did not evidently affect body weight in hyperlipidemic rabbits. All these results suggest that polydatin from *Polygonum cuspidatum* has favorable potency to develop a hypolipemic and/or hepatoprotective agent in clinic. However the mechanism of hypolipemic action of polydatin is in need of further clarity.

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1. Introduction

Lipids include cholesterol, cholesterol esters, phospholipids and triglycerides. Hyperlipidemia is an elevation of lipids (fats) in the bloodstream and usually means high cholesterol and high triglyceride levels. High lipid levels can harden the arteries or speed up the process of atherosclerosis, which increases the risk of heart disease, stroke, and other vascular diseases. Therefore, there are numerous chemical agents sought as lipid-lowering medication. They have played an important role in managing many symptoms in medical illness. However, the toxic effects of these drugs have also been established,

including various withdrawal syndromes, myopathy, stomach upset, rubefaction, tickle, liver dysfunction. Even Niacin (vitamin B3) also has adverse effect on glycemia [1–4]. There are numerous herbal medicines exerting good hypolipidemic actions with few side effects in Asian countries, especially in China, Japan and Korea, such as *Rhizoma Polygonati Odorati* [5], *Picrorrhiza kurroa* [6], *Phyllanthus niruri* [7], *Cercipia obtusifolia* [8], and so on.

Rhizoma Polygoni Cuspidati (RPC), the root and rhizome of *Polygonum cuspidatum*, is a widely used traditional medicine in Asia which has been shown to possess antibacterial [9], antioxidant [10] and anti-angiogenesis [11] activities. The ethanol extracts of RPC could evidently attenuate TNF-mRNA levels in adipose tissue, and decrease the contents of triglyceride, total cholesterol, and glucose in liver tissue in mice [12], which indicates that RPC may contain active ingredients characterized by lipid-lowering potential in serum.

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However, there are numerous constituents existing in RPC, such as emodin, chrysophanol, rhein, resveratrol, polydatin, catechin, quercetin, and so on [9]. The exact constituents possessing hypolipidemic activity remain incompletely understood.

Polydatin, the principal constituent in RPC, is an important stilbenoid and possesses several activities proved by previous studies: It exerts a neuroprotective effect on cerebral injury induced by ischemia/reperfusion [13], promotes weight loss and enhances diet attitudes in low-income mothers of young children [14], and protects the primarily cultured rat hepatocytes against CCl₄-induced injury [15]. Recent study also reported that it could reduce lipid oxidation [10]. Nevertheless, to our knowledge, there have been no reports yet on the effects of polydatin on lipid profile in hyperlipidemic rabbits.

In the present study, the hyperlipidemic rabbit model was employed, and the parameters of hyperlipidemia, including total cholesterol (TC), triglyceride (TG), low-density lipoprotein cholesterol (LDL-C) and high-density lipoprotein cholesterol (HDL-C), were investigated. Furthermore, the body weight, liver coefficient and cardiovascular index were also estimated. The aim of present study is to assess whether polydatin from *Polygonum cuspidatum* possesses the hypolipidemic activity or not.

2. Materials and methods

2.1. Polydatin

Polydatin was isolated and purified from the root and rhizome of *Polygonum cuspidatum* by Department of Pharmacognosy, School of Traditional Chinese Materia Medica, Shenyang Pharmaceutical University (Shenyang, China) and appeared as white powder. (Molecular formula C₂₀H₂₂O₈; molecular weight: 390.40; purity: 91.3%).

2.2. Animals and grouping

Forty male Japanese Giant Ear Rabbits, purchased from Shanghai Si-Lai-Ke Experimental Animal Ltd. (Shanghai, China) with an initial body weight of 1.8–2.0 kg, were used in this study. They were housed individually in a regulated environment (20 ± 2 °C), with a 12-h light/dark cycle (lights on: 08:00–20:00 h). Food and water were given ad libitum throughout the experiment. All animal treatments were strictly in accordance with international ethical guidelines and the National Institutes of Health Guide concerning the Care and Use of Laboratory Animals, and the experiments were carried out with the approval of the Committee of Experimental Animal Administration of the University.

After 2-week acclimation, animals were randomly divided into five groups ($n = 8$): one normal group, one control group and three groups treated with polydatin (25, 50, and 100 mg kg⁻¹ per day). The animals in normal group were fed a basic diet and other groups of animals fed a high fat/cholesterol diet for 6 weeks. Dietary ingredients were listed in Table 1.

Table 1
Composition of the basic diet and high fat/cholesterol diet (%)

Ingredients	Basic diet	High fat/cholesterol diet
Corn	35	27
Wheat	25	20
Wheat bran	35	27
Fish meal	2	2
Bone meal	2	2
Salt	1	1
Yolk	0	15
Cholesterol	0	1
Lard	0	5

2.3. Administration

Polydatin, isolated from the root and rhizome of *Polygonum cuspidatum*, was dissolved in distilled water prior to administration. Three weeks after a high fat/cholesterol diet, three treatment groups of animals were orally administered polydatin (25, 50, and 100 mg kg⁻¹ per day) by intubation for 3 weeks. Other groups of animals were orally administered distilled water of same volume, and they were run concurrently with polydatin treated groups. Polydatin was given in a volume of 10 ml kg⁻¹ irrespective of dose.

2.4. Body weight

Body weight was measured at weeks 0, 3 and 6 after experiment. To reduce the error originating from feeding, all animals were fasted (water was not restricted) for 12 h before measurement.

2.5. Liver coefficient

At the end of 6-week feeding period, the diet was removed 12 h prior to anesthetizing the animals with pentobarbital (30 mg kg⁻¹) injected into marginal ear vein. At sacrifice, livers were rapidly removed, washed in cold 0.9% normal saline, blotted onto filter-paper and weighed. The liver weight

Table 2
Effects of polydatin on body weight of rabbits

Group	Animals (N)	Dose (mg/kg d ⁻¹)	0 week (kg)	3 weeks (kg)	6 weeks (kg)
Normal	8	Distilled water	2.13 ± 0.14	2.46 ± 0.26	2.76 ± 0.34
Control	8	Distilled water	2.10 ± 0.11	2.41 ± 0.27	2.59 ± 0.26
Polydatin	8	25	2.17 ± 0.18	2.48 ± 0.29	2.67 ± 0.27
Polydatin	8	50	2.09 ± 0.14	2.42 ± 0.22	2.55 ± 0.30
Polydatin	8	100	2.11 ± 0.16	2.44 ± 0.23	2.62 ± 0.33

When rabbits were fed a high fat/cholesterol diet for 3 or 6 weeks, their body weight showed a downward trend compared with that of animals fed a basic diet. Although the body weight in hyperlipidemic rabbits given polydatin (25, 50, and 100 mg kg⁻¹ po) still for 3 weeks displayed an upward trend, there was no significant difference detected among the groups throughout the whole experiment.

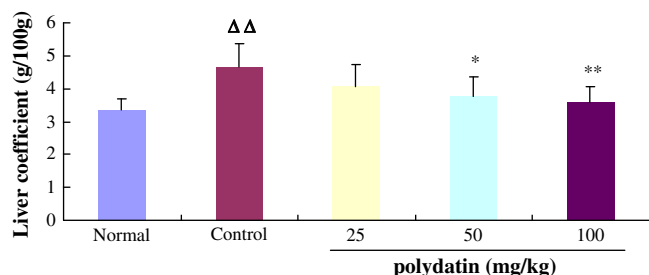


Fig. 1. Effects of polydatin on liver coefficient of rabbits (g/100 g). In control group of rabbits fed a high fat/cholesterol diet for 6 weeks, the liver coefficient was significantly higher compared with that of normal group. However, the liver coefficient markedly decreased in groups treated with polydatin (50 and 100 mg kg⁻¹ still for 3 weeks) compared with control group. ΔΔ $P < 0.01$ compared with normal rabbits; * $P < 0.05$, ** $P < 0.01$ compared with control rabbits. Data are expressed as the mean $\pm$ SD.

was detected as g levels and regarded as absolute weight. In order to reduce effects of individual body weight difference on liver weight, liver coefficient was calculated as the relative liver weight.

$$\text{coefficient}(\%) = \text{liver weight} / \text{weight at sacrifice} \times 100$$

2.6. Serum lipid assays

Blood samples were collected from marginal ear vein of overnight fasted rabbits on days 0, 21 and 42 after treatment. The blood was put at room temperature for 1 h; serum was then separated by centrifugation at $1500 \times g$ for 15 min.

The serum was kept under nitrogen in glass tubes to prevent oxidative degradation. The levels of serum TC, TG and HDL-C were measured by colorimetric enzymatic methods using commercial kits purchased from Nanjing Jiancheng Bioengineering Institute (Nanjing, China) [16].

2.7. Statistical analysis

The data were analyzed using a SPSS 11.0 statistical package. The data for multiple comparisons were performed by one-way ANOVA followed by LSD t -test. A value of $P < 0.05$ was considered statistically significant and all results are presented as the mean $\pm$ SD.

3. Results

3.1. Effects of polydatin on the body weight

As the animals were fed high fat/cholesterol diet for 3 or 6 weeks, their body weight showed a downward trend compared with that of animals fed a basic diet. But there was no significant difference detected in body weight among the groups throughout the whole experiment (Table 2). The fact of similar changes in body weight between normal group and high fat/cholesterol diet treated group indicated that high fat/cholesterol diet influenced little on the body weight.

3.2. Effects of polydatin on the liver coefficient

As compared with normal group of animals, the liver coefficient significantly increased in control group of animals fed high fat/cholesterol diet for 6 weeks. While 3 weeks after administration of polydatin, the liver coefficient significantly declined in both middle dose group and high dose group compared with that of control group of animals. However, low dose of polydatin (25 mg/kg d⁻¹) had no significant influence on liver weight in rabbits fed high fat/cholesterol diet (Fig. 1).

3.3. Effects of polydatin on the serum levels of TC, TG, HDL-C and LDL-C

As can be seen in Tables 3–6, the serum levels of TC, TG, HDL-C and LDL-C significantly increased in animals fed with high fat/cholesterol for 3 and 6 weeks when compared with those in the normal group of rabbits fed a basic diet. When the rabbits were given polydatin at dosages observed for 3 weeks, the levels of TC, TG and LDL-C were significantly decreased except HDL-C compared with those observed in the control group of animals, this difference being statistically significant.

Although there was no statistically significant difference in the serum HDL-C levels, the ratio of TC to HDL standing for cardiovascular risk markedly decreased in polydatin-treated groups of animals when compared with the values observed in the control group. On the other hand, the ratio of TC to HDL was higher in the control group than in the normal group, this difference being statistically significant.

Table 3
Effects of polydatin on serum total cholesterol (TC) levels of rabbits

Group	Animals (N)	Dose (mg/kg d ⁻¹)	0 week (mmol/L)	3 weeks (mmol/L)	6 weeks (mmol/L)
Normal	8	Distilled water	1.37 $\pm$ 0.44	1.50 $\pm$ 0.62	1.35 $\pm$ 0.51
Control	8	Distilled water	1.43 $\pm$ 0.53	23.92 $\pm$ 4.47 ΔΔ	30.45 $\pm$ 5.39 ΔΔ
Polydatin	8	25	1.47 $\pm$ 0.65	26.21 $\pm$ 6.37 ΔΔ	24.37 $\pm$ 4.21*
Polydatin	8	50	1.63 $\pm$ 0.64	25.13 $\pm$ 5.31 ΔΔ	20.52 $\pm$ 4.88**
Polydatin	8	100	1.52 $\pm$ 0.55	24.79 $\pm$ 6.33 ΔΔ	16.61 $\pm$ 7.13**

All groups of animals fed a high fat/cholesterol diet for 3 or 6 weeks showed a significant increase in TC levels compared with normal group. After the next 3 weeks of treatment with polydatin (25, 50, and 100 mg kg⁻¹), the TC levels markedly decreased compared with control group. ΔΔ $P < 0.01$ compared with normal rabbits; * $P < 0.05$, ** $P < 0.01$ compared with control rabbits. Data are expressed as the mean $\pm$ SD.

Table 4
Effects of polydatin on high-density lipoprotein cholesterol (HDL-C) levels of rabbits

Group	Animals (N)	Dose (mg/kg d ⁻¹)	0 week (mmol/L)	3 weeks (mmol/L)	6 weeks (mmol/L)	TC/HDL (ratio)
Normal	8	Distilled water	0.39 ± 0.19	0.43 ± 0.27	0.33 ± 0.18	4.06 ± 0.41
Control	8	Distilled water	0.41 ± 0.22	4.44 ± 0.32 $\Delta\Delta$	5.23 ± 0.61 $\Delta\Delta$	5.64 ± 0.79 $\Delta\Delta$
Polydatin	8	25	0.36 ± 0.16	4.89 ± 1.05 $\Delta\Delta$	5.53 ± 1.44	4.49 ± 1.01*
Polydatin	8	50	0.38 ± 0.17	4.33 ± 0.93 $\Delta\Delta$	4.74 ± 1.26	4.27 ± 0.78**
Polydatin	8	100	0.40 ± 0.23	4.13 ± 0.74 $\Delta\Delta$	4.46 ± 1.58	3.73 ± 0.66**

All groups of animals fed high fat/cholesterol diet for 3 or 6 weeks showed a significant increase in HDL-C levels compared with normal group. After the next 3 weeks of treatment with polydatin (25, 50, and 100 mg kg⁻¹), the HDL-C levels had no significant changes compared with control group. But polydatin significantly decreased the ratio of TC to HDL in all dosage group compared with control group. $\Delta P < 0.05$, $\Delta\Delta P < 0.01$ compared with normal rabbits; * $P < 0.05$, ** $P < 0.01$ compared with control rabbits. Data are expressed as the mean ± SD.

4. Discussion

Hyperlipidemia has been already documented as one of the vital coronary risk factors [17]. Hyperlipidemia, especially hypercholesterol, is positively related to morbidity and mortality of coronary heart disease. Now in China, the morbidity of cerebrovascular and angiocardopathy disease is up to 8% and its mortality is almost 50%, making hyperlipidemia a leading cause of death. Therefore, the present study is catering for the urgency of recent situation in medicine.

The present experiment showed that, after a 2-week acclimation, there was no significant difference in animals. However, animals fed a high fat/cholesterol diet showed such symptoms of hyperlipidemia based on Chinese medicine theory as inappetent, accidie and diarrhea. On the other hand, it was also showed in Tables 3 and 6 that the serum lipid indexes, TC and TG, were higher in the control group than in the normal group, this difference being statistically significant. All these results showed that the rabbit hyperlipidemia model was established successfully by means of feeding a high fat/cholesterol diet over 3 weeks.

During whole experimental period the body weight of rabbits changed little (Table 2), which indicated that high fat/cholesterol diet did not evidently exert an influence on body weight of rabbits. But some previous studies showed that high fat/cholesterol diet could lead to an increase of body weight of mice [6,18]. This diversity probably resulted from different species of experimental animals used [19]. Rabbits are nearly entirely vegetarian and high fat/cholesterol intake can influence the serum lipid profile but not always improve the body fat [20]. On the other hand, little change in body weight also indicated that a high fat/cholesterol diet had no effect on the normal growth and development of rabbits and the

method of establishing hyperlipidemic rabbit model was credible.

In the present research, the liver weight was also considered. None-alcoholic fatty liver disease induced by hyperlipidemia is a kind of disorder in fat metabolism, which is harmful to the liver and a major cause of obesity-related morbidity and mortality. It was reported that some materials possessing hypolipemic properties also showed hepatoprotective effects especially in herbal extracts [21–23]. In this study, the liver coefficient significantly increased in control group of animals fed high fat/cholesterol diet for 6 weeks, but evident decline was found in the groups of animals given polydatin for 3 weeks. These results indicated that polydatin was characterized by an ameliorating effect on none-alcoholic fatty liver.

It is well-known that cholesterol is a fatty substance which is an important part of the membrane of cells in the body of animals. The liver is capable of removing cholesterol from the blood circulation as well as manufacturing cholesterol and secreting cholesterol into the blood circulation. TC is a measure of the total amount of all cholesterol in blood at a given time and is the sum of HDL-C, LDL-C, very low-density lipoprotein cholesterol (VLDL-C), and intermediate density cholesterol (IDL). As shown in Table 3, at weeks 3 and 6, the levels of TC significantly increased in the control group of rabbits. When hyperlipidemic rabbits were given polydatin at dosages observed for 3 weeks, the levels of TC evidently decreased.

HDL-C removes excess cholesterol from arteries and moves it to liver for further processing or to be eliminated from the body. The higher serum HDL-C is the better. Therefore the HDL-C is called “good” cholesterol. It also plays a key role in the protection against oxidative damage of membrane [24,25]. LDL-C contributes to buildup of fat deposits in the arteries (atherosclerosis), which can cause decreased blood

Table 5
Effects of polydatin on low-density lipoprotein cholesterol (LDL-C) levels of rabbits

Group	Animals (N)	Dose (mg/kg d ⁻¹)	0 week (mmol/L)	3 weeks (mmol/L)	6 weeks (mmol/L)
Normal	8	Distilled water	0.71 ± 0.41	0.67 ± 0.35	0.79 ± 0.48
Control	8	Distilled water	0.69 ± 0.32	9.21 ± 0.55 $\Delta\Delta$	11.76 ± 0.71 $\Delta\Delta$
Polydatin	8	25	0.75 ± 0.29	9.23 ± 0.68 $\Delta\Delta$	8.60 ± 1.54**
Polydatin	8	50	0.66 ± 0.27	9.76 ± 0.88 $\Delta\Delta$	8.24 ± 1.72**
Polydatin	8	100	0.74 ± 0.34	10.01 ± 0.93 $\Delta\Delta$	7.88 ± 1.58**

Compared with normal group, the LDL-C levels of rabbits fed high fat/cholesterol diet were significantly higher. But the levels of LDL-C evidently decreased after 3 weeks of treatment with polydatin (25, 50, and 100 mg kg⁻¹) compared with control group. $\Delta\Delta P < 0.01$ compared with normal rabbits; ** $P < 0.01$ compared with control rabbits. Data are expressed as the mean ± SD.

Table 6
Effects of polydatin on serum triglyceride (TG) levels of rabbits

Group	Animals (N)	Dose (mg/kg d ⁻¹)	0 week (mmol/L)	3 weeks (mmol/L)	6 weeks (mmol/L)
Normal	8	Distilled water	1.21 ± 0.32	1.18 ± 0.23	1.32 ± 0.48
Control	8	Distilled water	1.12 ± 0.36	4.72 ± 1.07	6.37 ± 2.10
Polydatin	8	25	1.25 ± 0.25	4.37 ± 1.36 _{△△}	5.33 ± 2.44
Polydatin	8	50	1.11 ± 0.24	4.58 ± 1.40 _{△△}	3.43 ± 2.26*
Polydatin	8	100	1.29 ± 0.31	4.64 ± 0.92 _{△△}	2.37 ± 1.58**

The TG levels were significantly higher in rabbits fed a high fat/cholesterol diet for 3 or 6 weeks than normal group. But it was markedly reduced by administration of polydatin (50 and 100 mg kg⁻¹) for 3 weeks compared with control group except at low dose Group (25 mg/kg d⁻¹). _{△△}P < 0.01 compared with normal rabbits; *P < 0.05, **P < 0.01 compared with control rabbits. Data are expressed as the mean ± SD.

flow and head attack. So it is always called “bad” cholesterol, and a less levels are desirable.

As to the data of HDL-C in the experiment (Table 4), there was also a remarkable increase of the serum HDL-C levels in the rabbits fed a high fat/cholesterol diet for 3 weeks and it changed little after the next 3 weeks of treatment with polydatin. The increase of “good” cholesterol does not mean high fat/cholesterol diet is favorable to serum lipid profile levels, because the increase of “good” probably does not surpass that of “bad”, which has negative effects on lipid-lowering. Therefore, the ratio of TC to HDL-C (TC/HDL) was introduced.

This ratio represents an atherogenic index, which is an important prognostic marker for cardiovascular disease [26–28]. Some healthcare professionals often use this ratio to assess the risk for developing heart disease. Lower ratio of TC to HDL-C means lower risk. An elevated TC/HDL-C ratio is usually associated with a low HDL-C and/or elevated TG. In this experiment, putting aside the confusing data of HDL-C levels, TC/HDL-C ratio reflected the true effect of polydatin, which could exactly reduce the cardiovascular risk. On the other hand, the administration of polydatin at doses observed for 3 weeks also decreased evidently the serum LDL-C levels in hyperlipidemic rabbits.

Triglycerides are composed of three fatty acids and glycerol. Like cholesterol, they circulate in blood, but are stored in body fat and used when body needs extra energy. Excess triglyceride in plasma is called hypertriglyceridemia, which is relative to the occurrence of coronary artery disease. In this study, TG levels significantly increased in the animals fed a high fat/cholesterol diet for 3 and 6 weeks, but markedly decreased after a 3-week treatment with polydatin.

In summary, the administration of polydatin could decrease the serum levels of TC, TG and LDL-C. The ratio of TC to HDL-C and the liver coefficient were also reduced. But both polydatin and high fat/cholesterol diet did not evidently affect body weight in hyperlipidemic rabbits. All these results suggest that polydatin from *Polygonum cuspidatum* has favorable potency to develop a hypolipemic and/or hepatoprotective agent in clinic. However the mechanism of hypolipemic action of polydatin is in need of further clarity.

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